

Cinematic 3D Portfolio + CMS

Everything about your portfolio website: what is built today, how it works (architecture & workflows), how to run, log in and deploy it, every section's current design with 6 alternative designs each, future sections, the phased roadmap, and the exact prompts to rebuild or upgrade it.



Owner: IIT Madras BS student · Built with Next.js 16 · Three.js · GSAP
Generated 1 October 2026 · Status: Phase 1 complete (demo content)

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01 • Summary & current status

A dark sci-fi, cinematic personal portfolio with an interactive 3D portrait card, a particle-galaxy background and scroll-driven animation — plus a private admin panel so you can change **every section, photo, certificate and project from the browser**, without touching code or redeploying.

Area	Status today	Notes
Public website (13 sections)	BUILT	Filled with demo content ("Your Name", placeholder art)
Admin CMS + login	BUILT	Single owner account, created by <code>npm run setup</code>
Security layer	BUILT	Rate limits, CSP, CSRF checks, upload validation, audit log
Error / empty / offline states	BUILT	404, 500, maintenance, offline, session-expired, no-results
Responsive design	VERIFIED	320 px → 1920 px, no horizontal scroll on any page
Tests & checks	PASSING	20 unit tests · typecheck · lint · production build
Your real content	PHASE 2	Send details (section 16) or add them in the admin
Hosting / domain	PHASE 3	Vercel + Turso + Vercel Blob (all have free tiers)
Privacy notice	PHASE 2	Template provided — needs your real details

About the "1 Issue" / hydration message you saw: it was caused by the **Grammarly** browser extension adding `data-gr-ext-installed` attributes to the page before React loaded — not by the website. It is now silenced with `suppressHydrationWarning` on `<body>` . While checking your live site I also found and fixed a real bug: in development the intro preloader could stay stuck at 000% (React Strict Mode ran its effect twice).

02 • Phases — stage 1 to stage 5

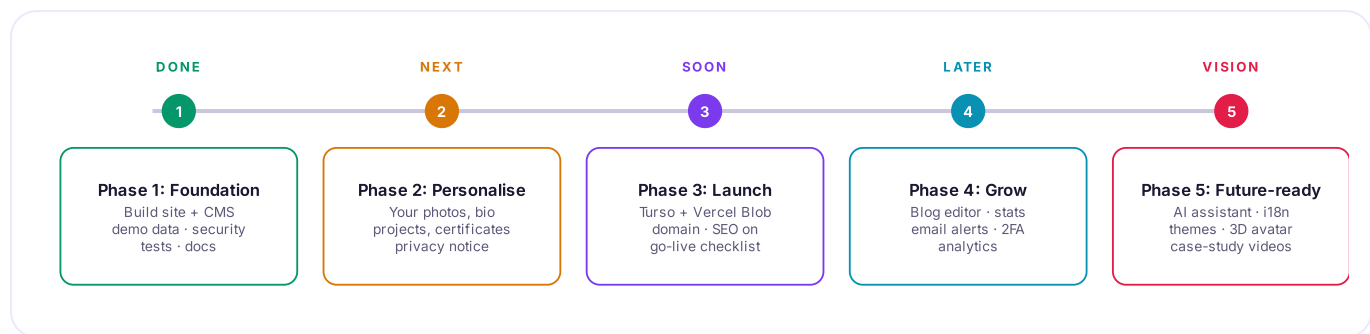


Figure 1 — Delivery phases. Phase 1 is complete; each later phase builds on the same codebase.

Phase	What happens	Outcome
1 • Foundation DONE	Plan, architecture, design system; 13 sections with 3D/motion; admin CMS; auth & security; error states; demo content; tests; docs	Working site on your computer (<code>npm run dev</code>)
2 • Personalise NEXT	Replace demo name, photos, bio, education, skills, projects, certificates; change admin password; write privacy notice; hide sections you don't need yet (Testimonials, Writing)	Your real portfolio locally; dashboard checklist all ✓
3 • Launch SOON	Create Turso DB + Vercel project + Vercel Blob; set env vars; seed cloud DB; custom domain; turn on search indexing; share link on LinkedIn/résumé	Live public URL
4 • Grow LATER	Built-in blog editor; GitHub/LeetCode live stats; email alerts for messages; privacy-friendly visit analytics; two-factor login; image cropper; automatic backups	Site that updates itself and tells you when people reach out
5 • Future-ready VISION	Theme presets (Netflix red, gold film, light); per-section layout switcher; 3D avatar (.glb); 'Ask me anything' AI assistant over your content; multi-language; recruiter view mode	A long-term personal brand platform through graduation and beyond

03 • Architecture

One Next.js application contains the public website, the admin panel and the API. A small **proxy** runs before every page to add security headers and guard `/admin`. Content lives in a database; photos and PDFs live in file storage. Locally both are plain files; in the cloud they are Turso and Vercel Blob — only environment variables change.

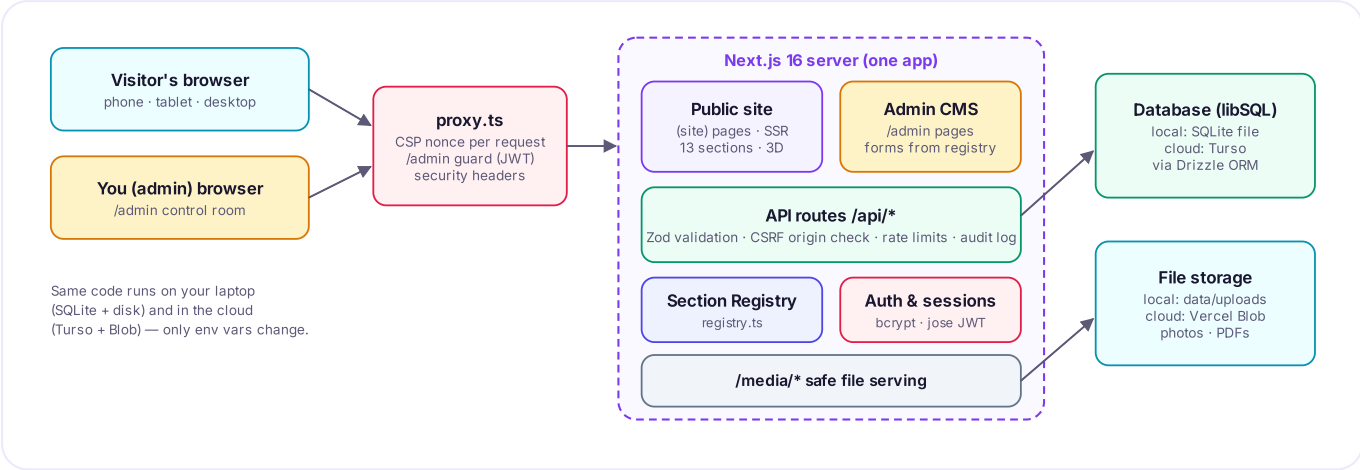


Figure 2 — System architecture.

Environment	Database	Uploads	Secrets
Your laptop (VS Code)	<code>file:./data/portfolio.db</code>	<code>data/uploads/</code>	<code>.env.local</code> (auto-created)
Vercel (recommended)	Turso <code>libsql://...</code>	Vercel Blob	Vercel → Settings → Environment Variables
Render / Railway / VPS	SQLite file on persistent disk (or Turso)	<code>data/uploads/</code> on disk	Host's env settings

04 • Workflows

A. A visitor opens your site

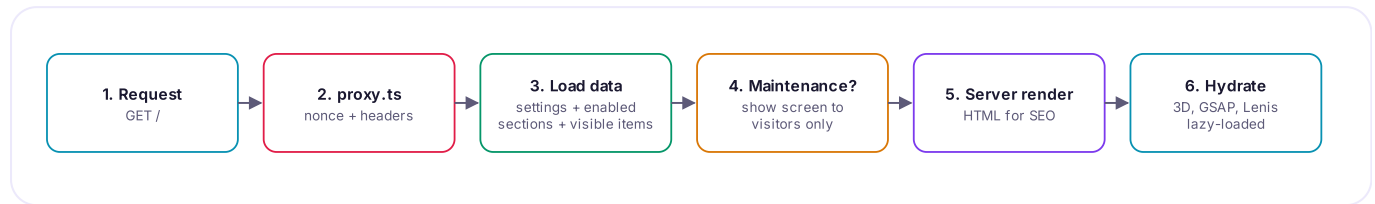


Figure 3 — Page request. Disabled sections and hidden entries are filtered on the server, so they never reach the browser.

B. You edit content in the admin

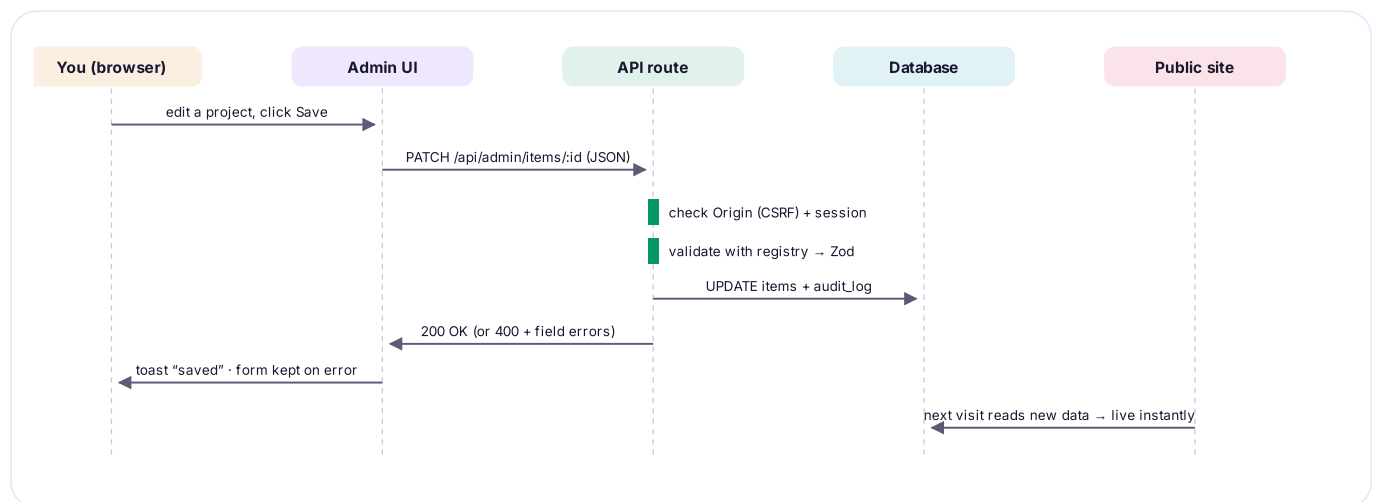


Figure 4 — Admin edit workflow. Changes are live on the next page load — no rebuild or redeploy.

C. Login & session security

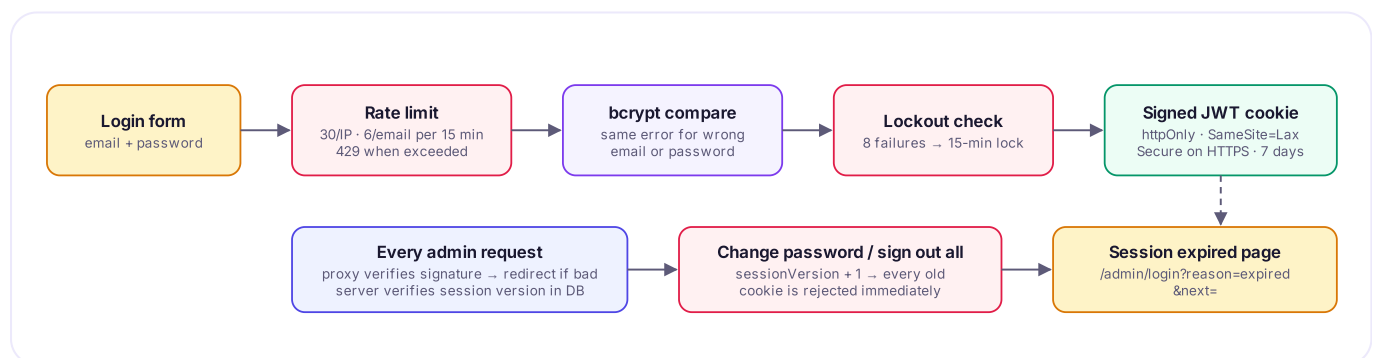


Figure 5 — Authentication and session lifecycle.

05 • Data model & the section registry

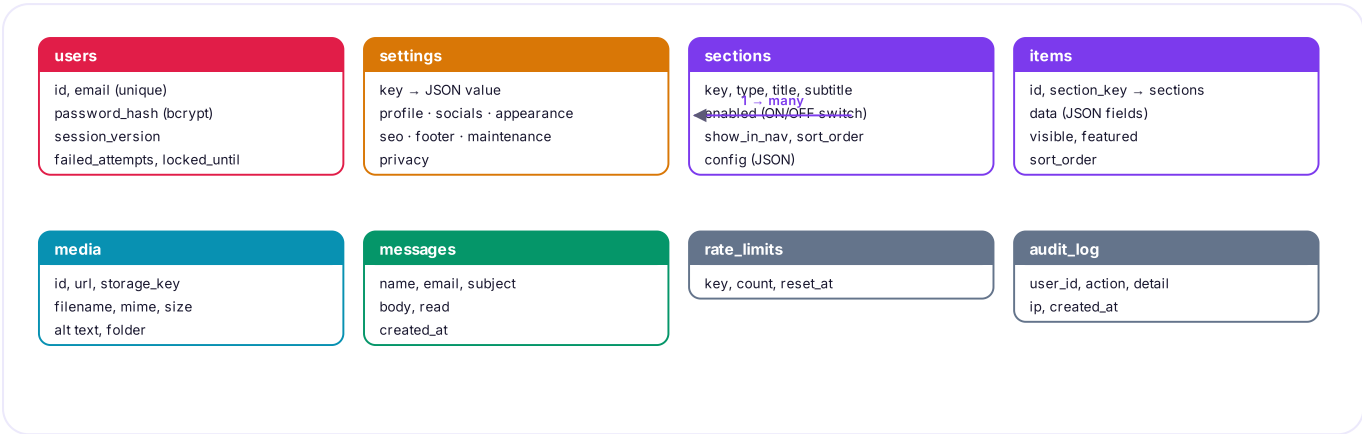


Figure 6 — Database tables (8). Section entries store their fields as JSON, so new fields need no database migration.

The **section registry** (`src/lib/registry.ts`) is the heart of the CMS. Every section type declares its fields once; the admin form, the server validation and the public page all read that single definition.

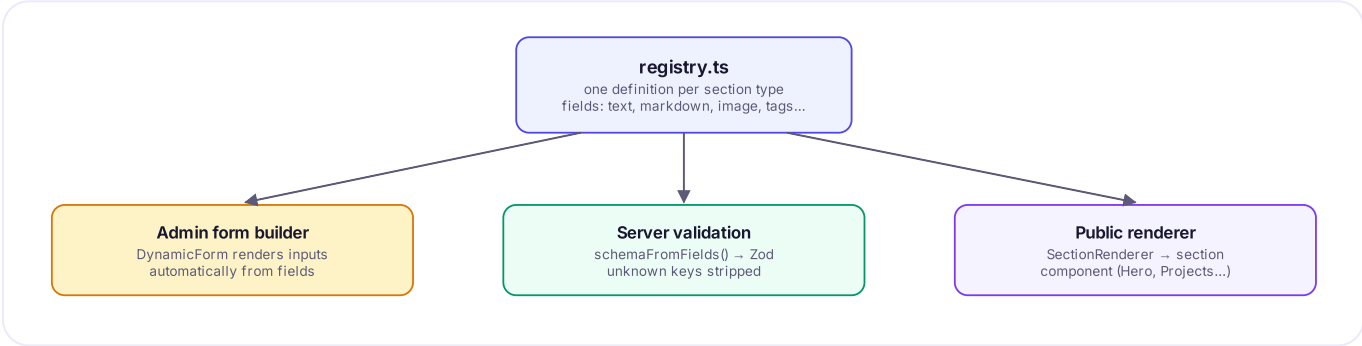


Figure 7 — Add a field in one place and it appears in the editor, is validated, and is available to the design.

06 • Tech stack

Layer	Technology	Why it was chosen
Framework	Next.js 16 (App Router) · React 19 · TypeScript	Server rendering for SEO & link previews, API in the same app, deploys anywhere
Styling	Tailwind CSS 4 + design tokens	Consistent spacing/colours; accent colours editable from admin
3D	Three.js · React Three Fiber · drei	3D portrait card, galaxy particles, orbit rings, sparkles
Motion	GSAP ScrollTrigger · Framer Motion · Lenis	Pinned film-reel, reveals, counters, smooth scrolling
Data	Drizzle ORM · libSQL (SQLite / Turso)	Zero-setup locally; free cloud database
Auth	bcrypt · jose (signed JWT cookie)	No third-party login needed; sessions can be revoked
Validation	Zod (generated from the registry)	Every input checked on the server
Files	Local disk or Vercel Blob	Works on a VPS and on serverless hosting
Quality	Vitest · ESLint · TypeScript · Playwright checks	Tests for security rules and helpers
Fonts	Space Grotesk · Inter · JetBrains Mono (self-hosted)	Fast, private, no Google Fonts requests

07 • Feature inventory — global

Cinematic preloader

Name reveal + counter, once per visit; skipped for reduced-motion and background tabs.

Galaxy background

12,000 (desktop) / 5,000 (mobile) particles reacting to mouse & scroll; dims after the hero.

Glow cursor

Dot + trailing ring that grows over links (desktop only).

Smooth scrolling

Lenis synced with GSAP; anchor links glide to sections.

Film grain & vignette

Subtle cinematic texture (toggle in admin).

Smart navbar

Glass on scroll, active-section highlight, 'More' menu, mobile drawer, résumé button.

Section on/off

Every section can be hidden, reordered, renamed, kept out of the menu.

Case-study pages

/projects/<slug> with video, gallery, tech sidebar, prev/next.

SEO

Title, description, share image, JSON-LD Person, sitemap, robots (indexing off until launch).

Maintenance mode

Visitors see a 'be right back' screen; you still see the full site.

Offline banner

Tells visitors when they lose connection.

Accessibility

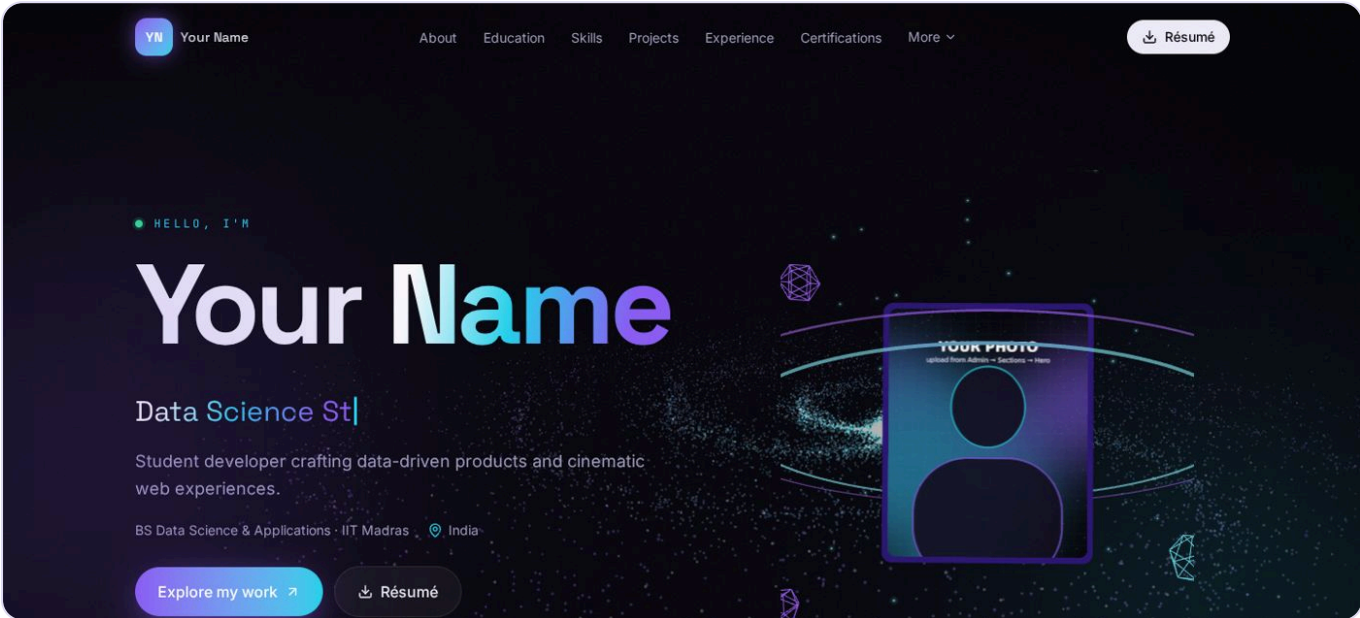
Skip link, labels, focus rings, keyboard dialogs, alt text, reduced motion.

SECTIONS

08 • Every section: today’s design + 6 future options

Each section below shows what it looks like now, what you can change from the admin, and six alternative designs you could switch to later. A future upgrade can add a **“Layout” selector** to any section so you can flip between designs from the admin (see the upgrade prompt in section 15).

08.01 Hero (3D intro)



Hero (3D intro) — current design (desktop, demo content)

Current design. Full-screen opening scene. Your name rises word-by-word with a violet → cyan gradient, roles type themselves out, and a floating holographic 3D card shows your photo (front) and a second photo (back). Visitors drag/swipe to spin it; it snaps back to face front. Orbit rings, wireframe crystals and sparkles float around it over the galaxy background.

You control (admin). Greeting · rotating roles · button label & link · résumé button on/off · 3D on/off · front/back photos · scroll hint.

Six alternative designs

A. Cinematic video hero

Muted looping showreel/video background with your name over a letterbox frame (like the friend's cinematic repo).

B. 3D avatar model

A rigged .glb avatar (Ready Player Me / Blender) that looks at the cursor and waves on load.

C. Netflix profile picker

“Who's watching?” screen with profiles: Recruiter · Developer · Student — each opens a tailored order of sections.

D. Terminal intro

A hacker-style terminal that types `whoami`, prints your stack, then ‘boots’ the site.

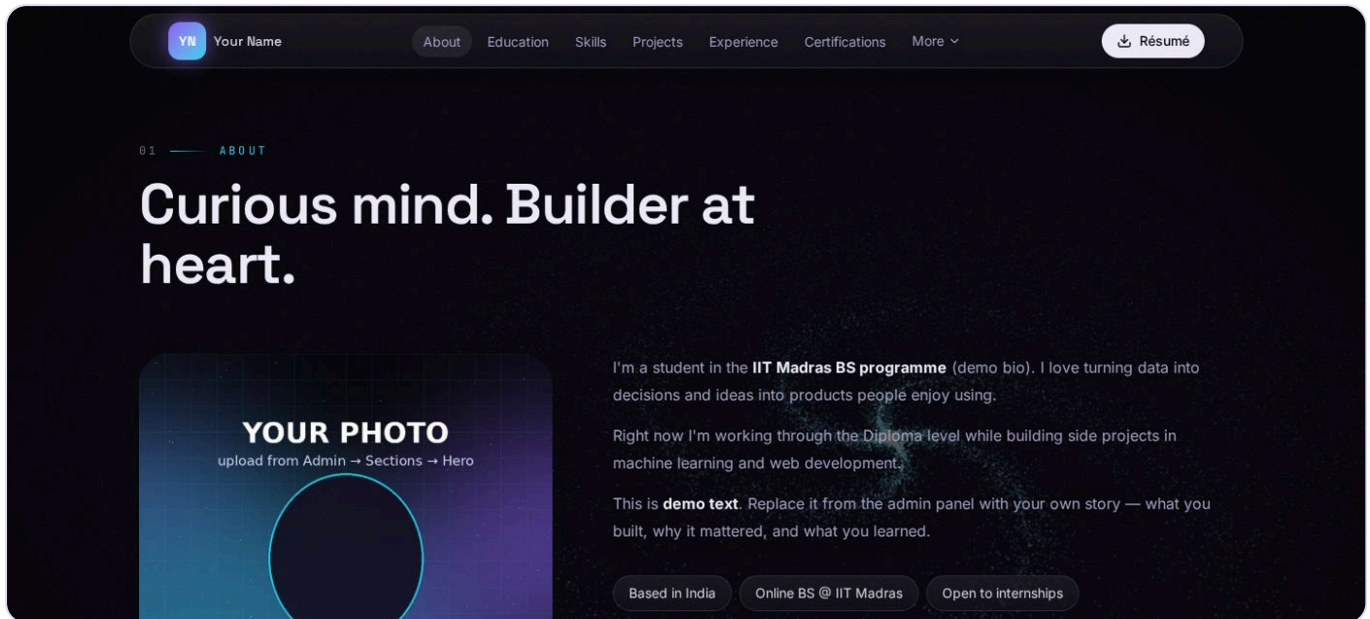
E. Split-screen parallax

Big portrait on one side, layered parallax text and floating skill icons on the other.

F. Particle-portrait

Your photo rebuilt from thousands of particles that scatter on hover and re-form.

08.02 About me



About me — current design (desktop, demo content)

Current design. Tilt-on-hover glass photo card with your name, a Markdown biography, quick-fact chips, four animated counters (count up when scrolled into view) and interest tags.

You control (admin). Heading · biography (Markdown) · main photo · extra photos · quick facts · interests · stats (label, number, suffix).

Six alternative designs

A. Story timeline

Your journey as chapters (school → diploma → IITM BS → future) revealed on scroll.

B. Bento grid

Apple-style tile grid: photo, bio, location map, now-playing, current course, fun fact.

C. Video introduction

A 60-second 'meet me' video with captions and a transcript.

D. Polaroid stack

Photos as a draggable polaroid pile with handwritten captions.

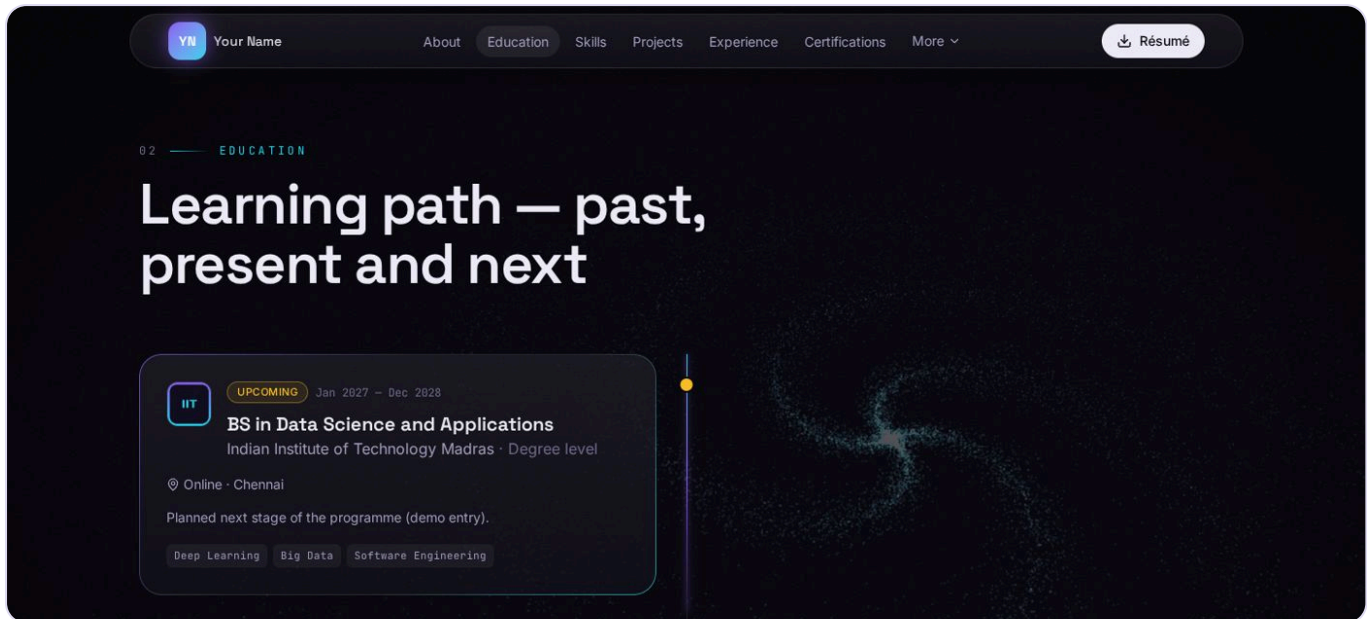
E. Interview Q&A

Accordion of 'What drives you?', 'Favourite project?', etc.

F. Day-in-the-life

Horizontal scroll of a typical study/coding day with icons and times.

08.03 Education



Education — current design (desktop, demo content)

Current design. Alternating left/right timeline whose glowing line ‘draws’ itself as you scroll. Each card shows a status badge (Completed / Ongoing / Upcoming — upcoming cards are dashed), dates, grade, location, logo, Markdown description and key-course chips.

You control (admin). Institution · degree · field · status · start/end · grade · location · logo · description · courses · link. Reorder, hide, add, delete.

Six alternative designs

A. Level-progress tracker

IITM-style ladder: Foundation → Diploma → BSc → BS with a progress bar showing where you are.

B. Course transcript table

Searchable table of every course with grade and term.

C. Semester cards carousel

One card per term with courses, GPA and highlights.

D. Skill-tree map

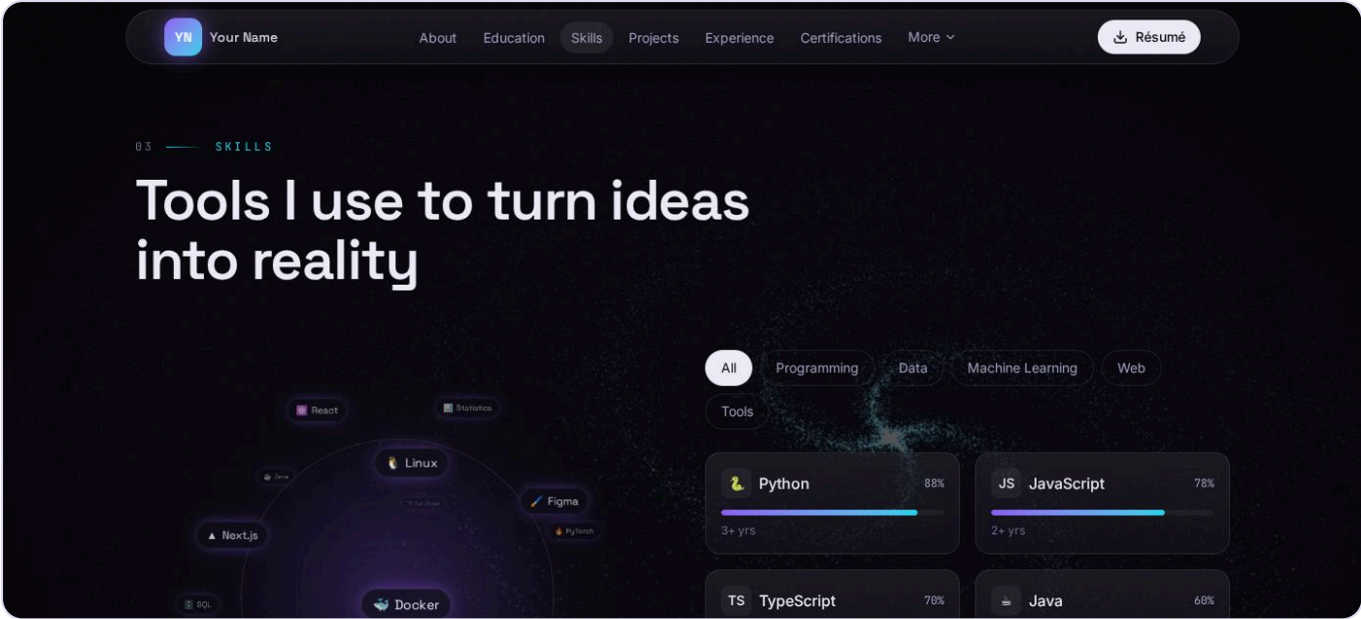
Courses as nodes in a game-like tree unlocking skills.

E. Academic stats dashboard

Charts of credits earned, CGPA trend and course categories.

F. Minimal list

Clean typographic list for a more formal, recruiter-first look.



Skills — current design (desktop, demo content)

Current design. A rotating 3D tag sphere (drag to steer) beside category tabs and animated proficiency bars with years of experience.

You control (admin). Skill name · category · level 0-100 · icon/emoji · years. Section toggles: sphere on/off, bars on/off.

Six alternative designs

A. Orbiting planets

Each category is a planet; skills orbit it in 3D.

B. Radar chart

Spider chart comparing categories (Programming, Data, ML, Web, Tools).

C. Logo marquee

Two infinite rows of tech logos scrolling in opposite directions.

D. Skill cards with projects

Each skill links to the projects where you used it.

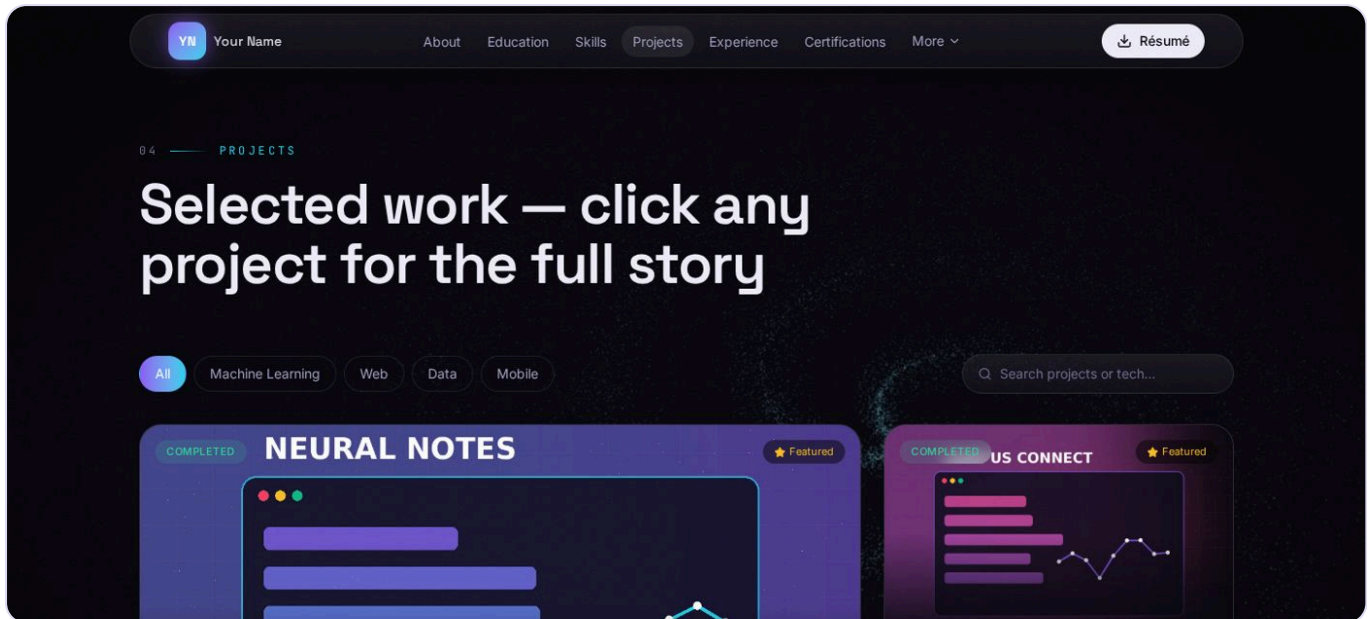
E. Heat-map grid

GitHub-style grid showing practice intensity per skill.

F. Certification-backed badges

Badges that show which certificate proves each skill.

08.05 Projects



Projects — current design (desktop, demo content)

Current design. Featured project spans two columns; cards tilt in 3D with a moving glare. Category filter chips, live search (with a 'no results' state), status badges (Completed / In progress / Coming soon), 'Show more', and a full case-study page per project with cover, video, Markdown story, screenshots, tech sidebar and previous/next navigation.

You control (admin). Title · slug · summary · status (incl. planned/future) · category · cover · gallery · case study · tech · role · dates · repo/live/video links · featured · visible. Section: filters, search, initial count.

Six alternative designs

A. Netflix rows

Horizontal rows per category with hover-expanding 'episode' cards and a trailer modal.

B. Scroll-stack cards

Full-width cards that stack and scale as you scroll (like the friend's ScrollStack).

C. 3D carousel / coverflow

Projects as a rotating 3D shelf.

D. Bento showcase

Mixed-size tiles with live previews / GIFs.

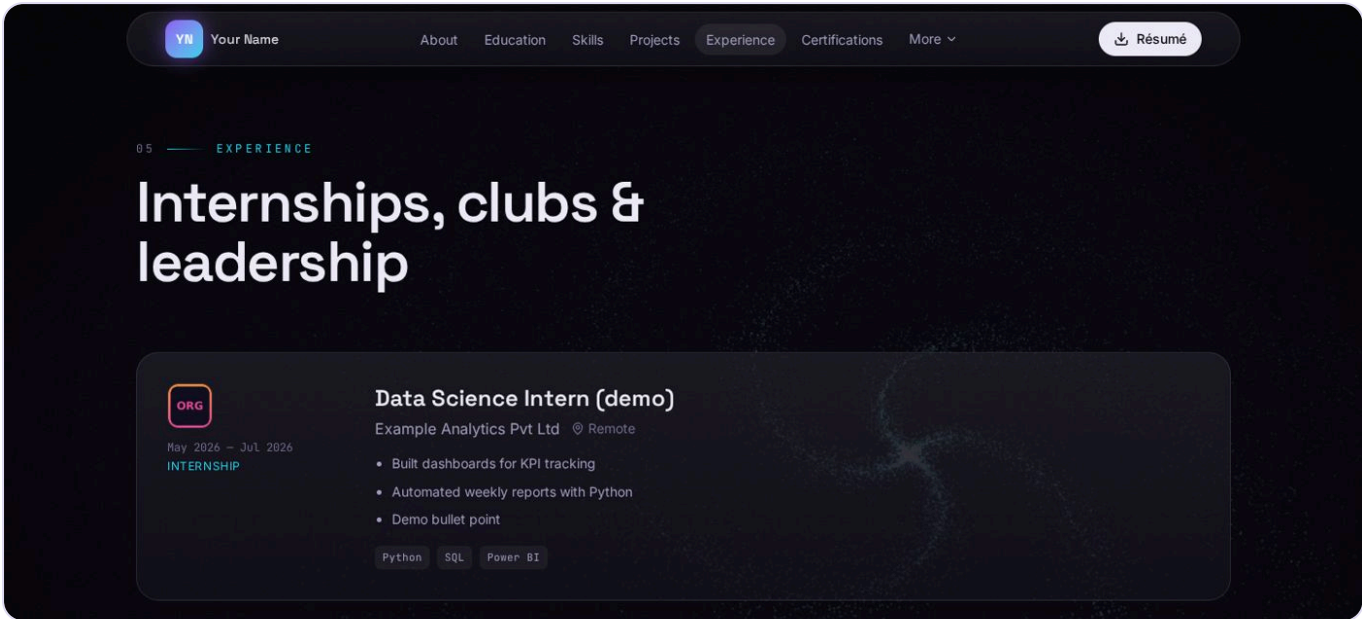
E. Timeline of builds

Projects on a chronological line with milestones.

F. Device mockups

Each project shown inside a laptop/phone mockup that rotates on scroll.

08.06 Experience



Experience — current design (desktop, demo content)

Current design. Glass cards with logo, dates and type label (Internship, Job, Freelance, Club, Volunteering, Leadership), role, organisation, location, Markdown bullet points and skill chips.

You control (admin). Role · organisation · type · dates (empty end = current) · location · logo · description · skills · link.

Six alternative designs

A. Vertical career timeline

Same timeline style as Education for a unified look.

B. Company logo wall

Logos that flip to reveal your role.

C. Impact metrics cards

Big numbers (e.g. '30% faster reports') as the hero of each card.

D. Tabs by type

Tabs: Internships · Clubs · Volunteering.

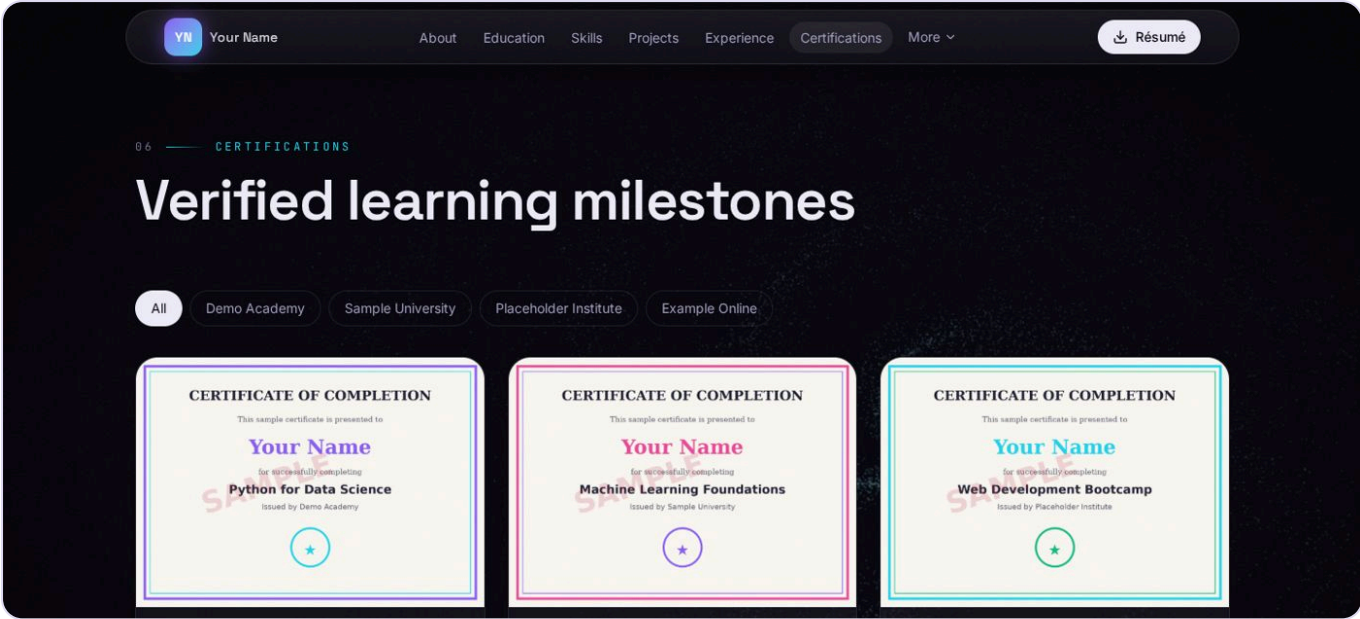
E. Map view

Pins on a world/India map for each role location.

F. Recommendation inline

Each role shows the matching testimonial underneath.

08.07 Certifications



Certifications — current design (desktop, demo content)

Current design. Tilt cards showing the uploaded certificate image; issuer filter chips; click opens a full-screen viewer with issuer, dates, credential ID, 'Verify' link and 'Open PDF'.

You control (admin). Title · issuer · issue/expiry dates · credential ID · verification URL · image · PDF · skills. Section: issuer filters on/off.

Six alternative designs

- A. Trophy shelf (3D)**

Certificates as framed plaques on a 3D wall you can pan.
- B. Badge grid**

Compact badge icons (Credly-style) with hover details.
- C. Timeline of learning**

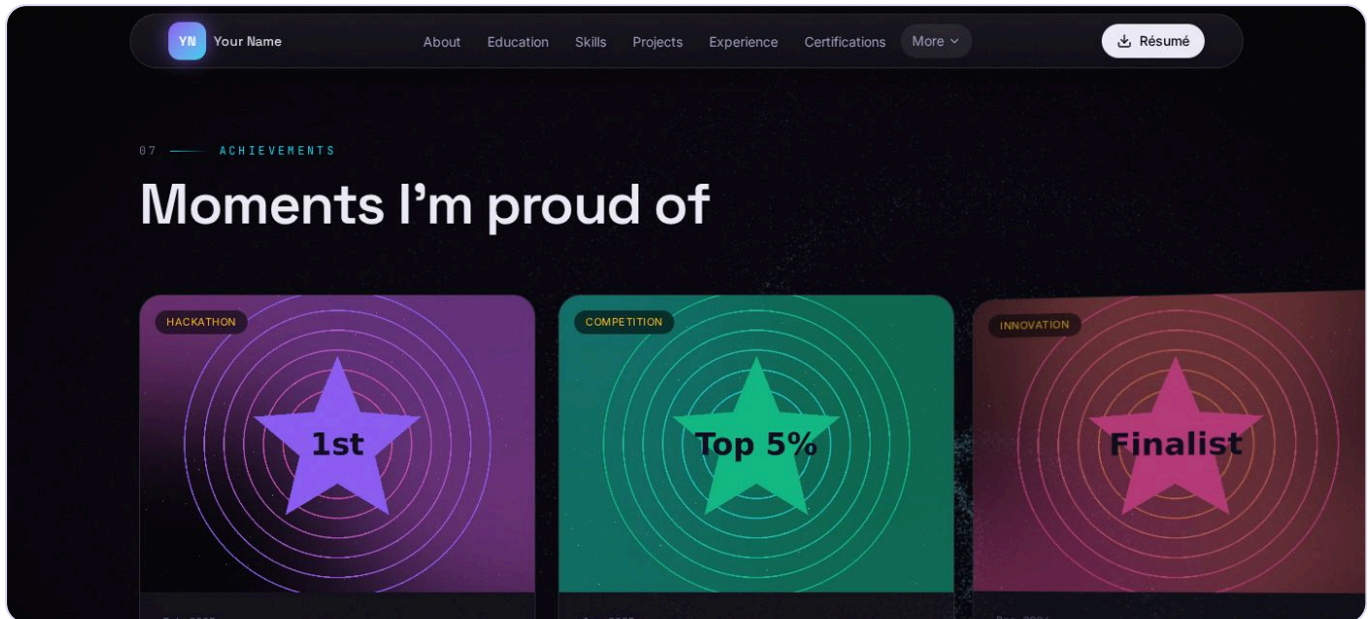
Certificates placed by date to show learning momentum.
- D. Grouped by skill**

Accordion per skill area.
- E. Swiper deck**

Swipeable card deck on mobile, overflow on desktop.
- F. Verification-first table**

Table with live 'verified ✓' status for recruiters.

08.08 Achievements & awards



Achievements & awards — current design (desktop, demo content)

Current design. Cinematic ‘film reel’: on desktop the section pins and vertical scrolling moves the cards sideways while each card rotates into view in 3D. On phones it becomes a swipeable row.

You control (admin). Title · organisation/event · date · category · description · image · link. Section: horizontal reel on/off (off = grid).

Six alternative designs

A. Medal podium

Top three achievements on a 3D podium, the rest listed below.

B. Count-up highlights

Large animated numbers (ranks, percentiles) with short captions.

C. Newspaper headlines

Each achievement styled as a press clipping.

D. Masonry wall

Photo-led masonry grid of event moments.

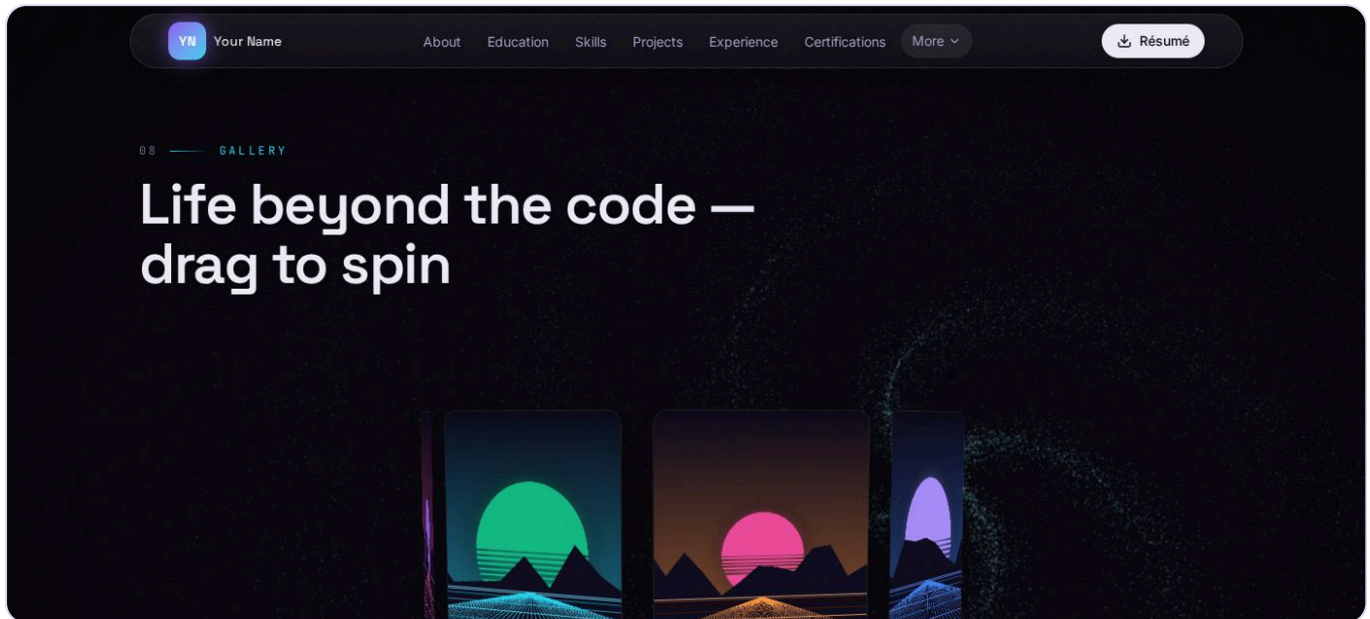
E. Timeline with photos

Year-by-year milestones.

F. Trophy cabinet

Glass cabinet with 3D trophies, click for details.

08.09 Gallery



Gallery — current design (desktop, demo content)

Current design. Draggable 3D ring carousel that slowly auto-rotates (arrow buttons for keyboard users) or a masonry grid; full-screen lightbox with caption, album, date and keyboard ←/→ navigation.

You control (admin). Photo · caption · album · date. Section: layout = 3D ring or masonry.

Six alternative designs

A. Album tabs

Filter by album (Events, Travel, Campus).

B. Infinite canvas

Drag around a huge canvas of photos in any direction.

C. Polaroid scatter

Photos thrown on a table; click to pick one up.

D. Before/after slider

For design or photography edits.

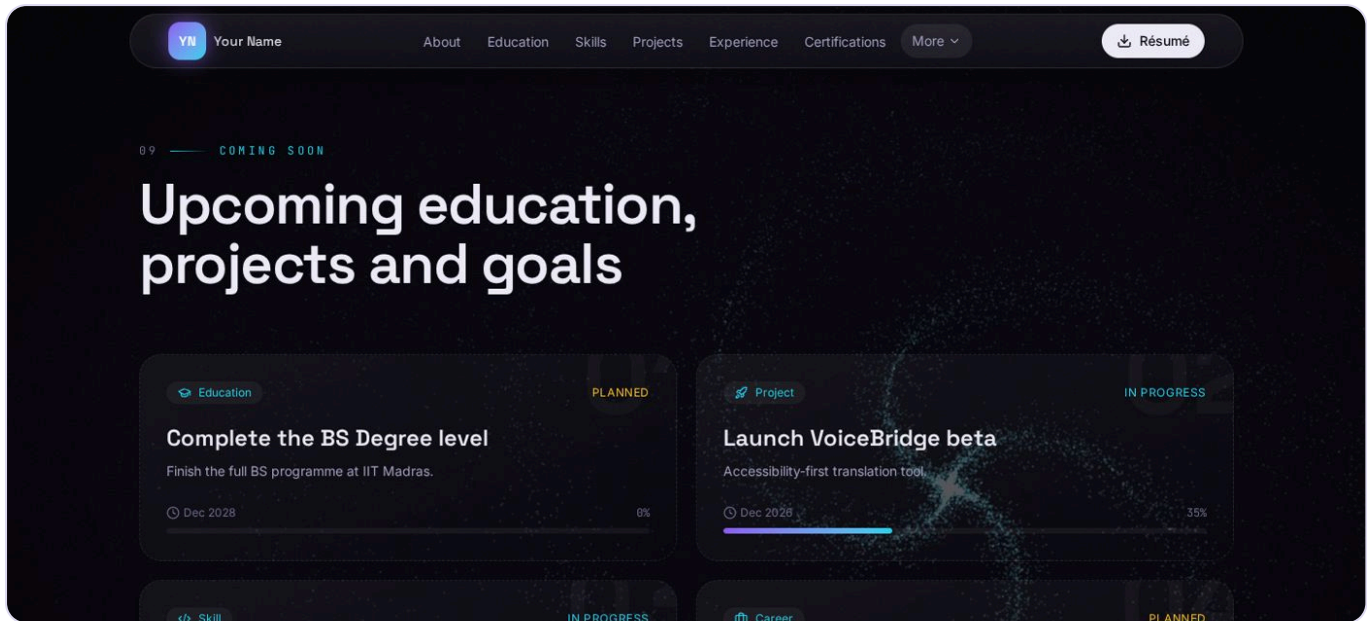
E. Story reels

Instagram-story style vertical swipes.

F. Video gallery

Short clips with hover-to-play.

08.10 Future & roadmap (Coming Soon)



Future & roadmap (Coming Soon) — current design (desktop, demo content)

Current design. 'Coming soon' trailer cards with large faded numbers, type tag (Education, Project, Skill, Career, Personal), status, target date and progress bar.

You control (admin). Title · type · target date · status · progress % · description.

Six alternative designs

A. Gantt-style timeline

Bars across months/years showing planned work.

B. Kanban board

Planned · In progress · Done columns.

C. Countdown cards

Live countdown to each target date.

D. Vision board

Image-led mood board of goals.

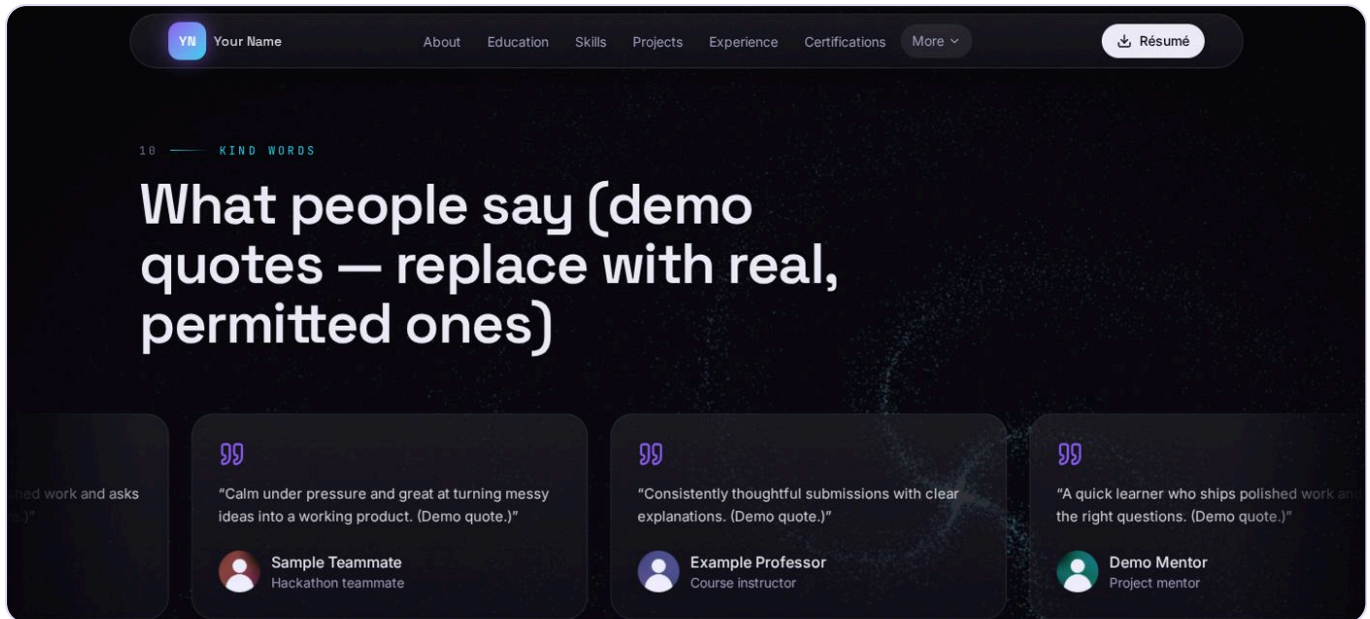
E. Quarterly OKRs

Objectives with key results and progress.

F. Learning path map

Path of courses/skills leading to a career goal.

08.11 Testimonials



Testimonials — current design (desktop, demo content)

Current design. Infinite marquee of quote cards with photo, name and role (pauses on hover). With fewer than three quotes it becomes a static grid.

You control (admin). Name · role · quote · photo · profile link. Only add real quotes you have permission to use.

Six alternative designs

A. Single spotlight slider

One large quote at a time with auto-advance.

B. Wall of love

Masonry of many short quotes.

C. Video testimonials

Short recorded clips.

D. LinkedIn embeds

Recommendations pulled from LinkedIn screenshots.

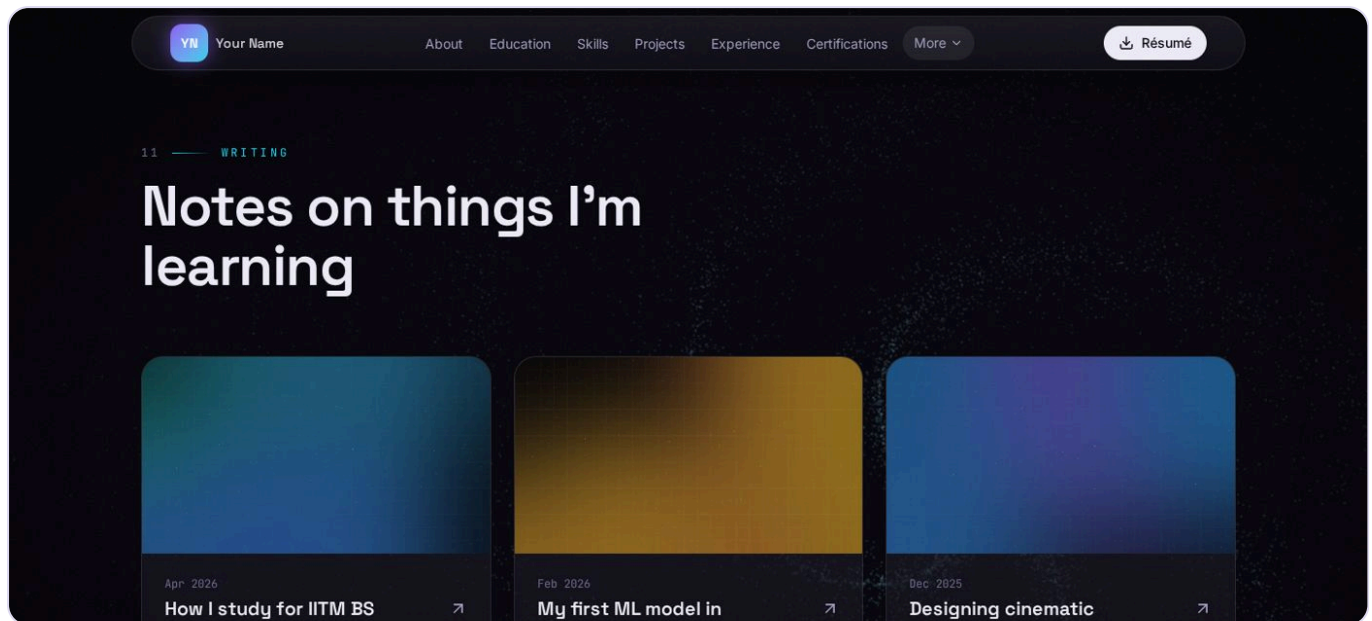
E. Rating + quote

Star ratings for freelance clients.

F. Quote over photo

Full-width quote with the person's photo blurred behind.

08.12 Writing / blog



Writing / blog — current design (desktop, demo content)

Current design. Cards with cover, date, title, excerpt and tags linking to external articles (Medium, LinkedIn, Hashnode).

You control (admin). Title · excerpt · URL · date · cover · tags.

Six alternative designs

A. Built-in blog

Write full posts in the admin with Markdown and their own pages (Phase 4).

B. Featured + list

One big featured article and a compact list.

C. Notes / TIL feed

Short 'today I learned' notes, Twitter-style.

D. Newsletter signup

Collect subscribers for updates.

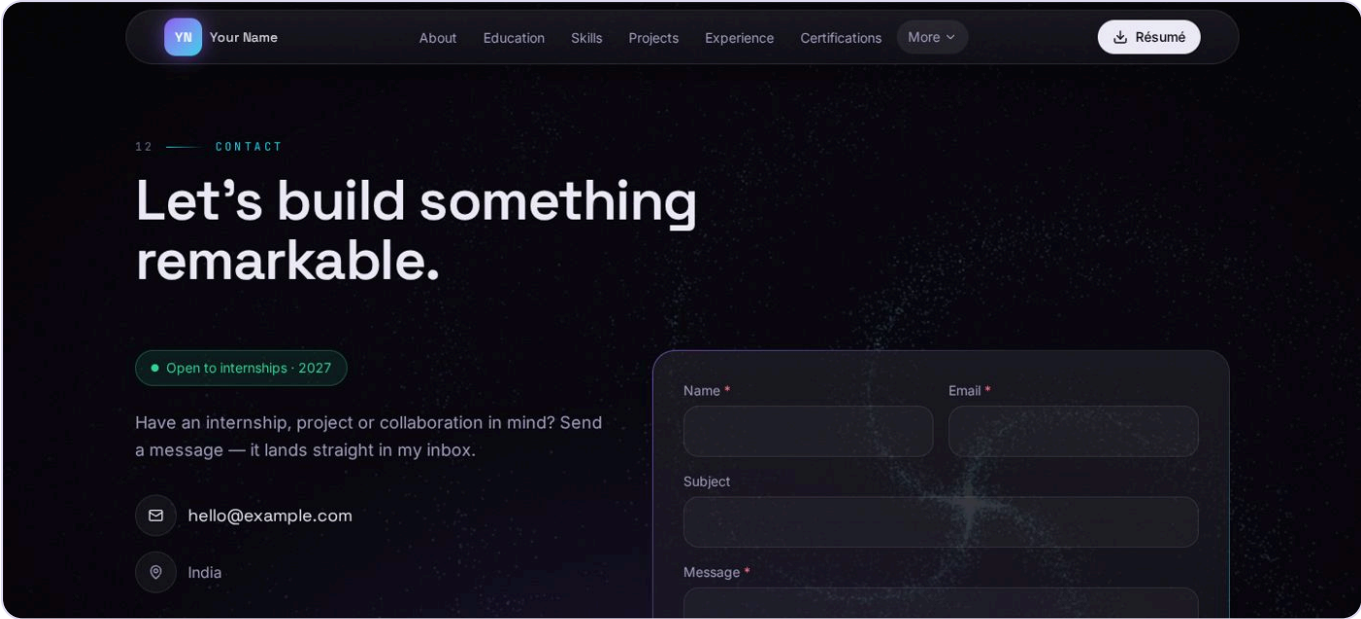
E. Talks & videos

Embedded YouTube talks and slides.

F. Reading list

Books/papers you're reading with mini reviews.

08.13 Contact



Contact — current design (desktop, demo content)

Current design. Availability badge, intro text, email, location, social icons and a secure form (validation, honeypot, rate limit). Messages land in the admin inbox; the form keeps your text if sending fails.

You control (admin). Heading · intro text · form on/off · show email on/off · availability badge.

Six alternative designs

A. Calendar booking

Embed Cal.com/Calendly to book a call.

B. Chat-style form

Questions asked one by one like a chat.

C. Email notification

Get an email (Resend) whenever someone writes — Phase 4.

D. Map + office hours

Location map and time-zone clock.

E. Quick links only

Minimal: big email + socials, no form.

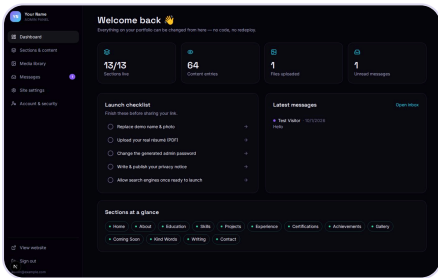
F. Recruiter pack

One click to download résumé + portfolio PDF.

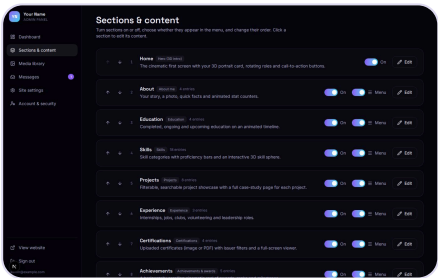
CONTROL ROOM

09 • Admin panel

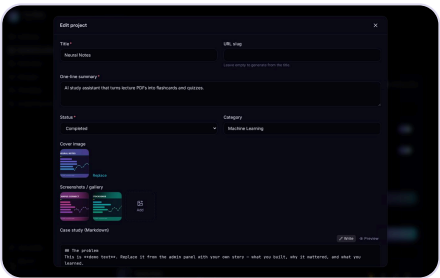
Page	What you can do
Dashboard	Live counts (sections, entries, files, unread messages), launch checklist, latest messages, sections at a glance
Sections & content	Turn sections on/off, show/hide in menu, move up/down, open editor
Section editor	Section heading & settings; add / edit / hide / feature / reorder / delete entries; unsaved-changes warning; preview link
Media library	Drag-and-drop photos & PDFs, automatic resize, alt text, search, filter, copy link, delete
Messages	Contact-form inbox, unread filter, reply by email, mark unread, delete
Site settings	Profile, résumé, social links, colours & effects, SEO, footer (incl. optional Admin link), maintenance, privacy notice
Account & security	Change password (strength meter), sign out of all devices, activity log with failed logins



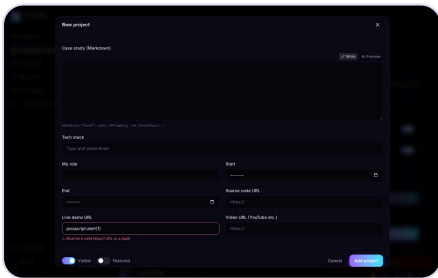
Dashboard



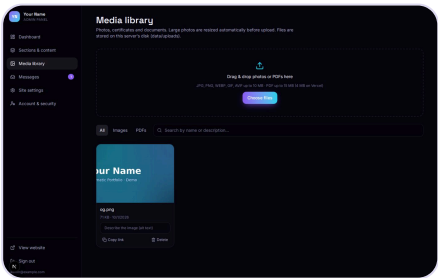
Sections manager — on/off switches and ordering



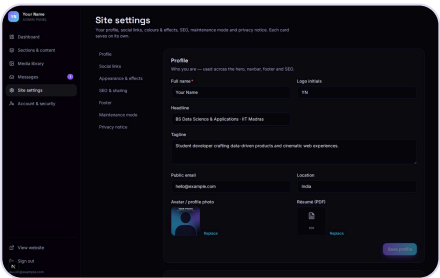
Entry editor generated from the registry



Validation keeps your input and highlights the problem



Media library



Site settings

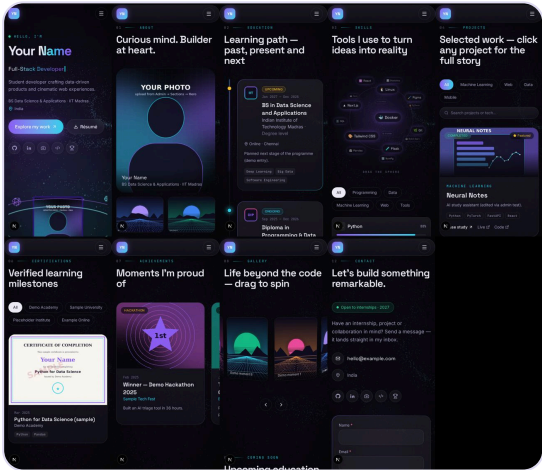
10 • Security & error handling

Risk	Protection
Password guessing	bcrypt (cost 12), 30 tries/IP and 6 tries/email per 15 min, account lock after 8 failures
Finding valid emails	Same message and same timing for wrong email and wrong password
Stolen / old sessions	httpOnly + SameSite cookie, Secure on HTTPS, 7-day expiry, 'sign out everywhere' and password change revoke all sessions
Bypassing the admin UI	Every admin API checks the session on the server, not only the page guard
Cross-site request forgery	Origin/Referer must match the site on every change request
Script injection (XSS)	Per-request nonce Content-Security-Policy, Markdown without raw HTML, link allow-list
Malicious uploads	File type detected from bytes, SVG/HTML refused, size limits, random names, sandboxed serving
Open redirects	After login you can only be sent to an /admin path
Clickjacking	frame-ancestors 'none' + X-Frame-Options
Contact spam	Honeypot field, minimum typing time, 5 messages/hour/IP
Leaking internals	Errors show a short reference ID; details only in server logs; secrets only in .env.local
State	What the user sees
404	'This scene doesn't exist' with Home and Projects buttons (real 404 status)
500 / crash	Friendly message, Try again button and a reference ID
Maintenance	'Will be right back' screen driven by the admin setting
Offline	Banner; forms keep what you typed
Session expired	Login page with explanation, returns you to the page you were on
Empty / no results	Explanation + next action (e.g. 'Clear search & filters')
Loading / success	Spinners and skeletons; toasts confirm the exact change

11 • Responsive & quality results

Every public and admin page was loaded at seven screen sizes and checked for horizontal overflow and broken layout.

Device size	Example	Result
320 × 640	Small Android / iPhone SE (1st gen)	✓ No overflow
375 × 667	iPhone SE / 8	✓
414 × 896	iPhone 11 / large Android	✓
768 × 1024	iPad portrait	✓
1024 × 768	iPad landscape / small laptop	✓
1280 × 800	Laptop	✓
1920 × 1080	Desktop monitor	✓



Phone view (390 px) — hero, about, education, skills, projects, certifications, achievements, gallery, contact

Check	Result
TypeScript	PASSED
ESLint	PASSED (0 problems)
Unit tests (20)	PASSED
Production build	PASSED
CSRF / auth / rate-limit / upload / maintenance checks	PASSED
Automated accessibility audit (axe)	Not run yet — recommended in Phase 4

12 • Login, passwords & deployment

Speed tip for your computer: your terminal ran the project from `/mnt/d/...` inside WSL, which is why `npm install` took 22 minutes and the first page took 100 s. Either run it from Windows PowerShell / the VS Code terminal in Windows mode, or copy the folder into your WSL home (e.g. `~/projects/cinematic-portfolio`) — it becomes 10–20× faster. Production hosting is not affected.

Is there a login button on the website?

Not by default — on purpose. Visitors and recruiters never see an admin button, which keeps the site clean and doesn't advertise the login page. You sign in by typing the address yourself:

- On your computer: `http://localhost:3000/admin` (redirects to `/admin/login`)
- After deployment: `https://your-domain/admin`

If you'd like a small link anyway: **Admin** → **Site settings** → **Footer** → “Show a small Admin sign-in link in the footer”.

Default email and password

There is no fixed default password (a shared default would be a security risk). `npm run setup` generates one the first time and prints it. Your current local login is:

- Email: `admin@example.com`
- Password: the one printed by your setup run (also stored as `ADMIN_PASSWORD` in `.env.local` in the project folder)

Change it after first login in **Account & security**. To use your own email, edit `ADMIN_EMAIL` and `ADMIN_PASSWORD` in `.env.local` *before* running setup on a fresh database.

Forgot the password?

```
npm run admin:reset-password -- admin@example.com "MyNewStrongPass2026"
```

After deployment (Vercel) — how the login works

1. On your computer, put the Turso URL and token into `.env.local` and set `ADMIN_EMAIL` / `ADMIN_PASSWORD` to what you want.
2. Run `npm run setup` — it creates the tables, content and **your admin account in the cloud database**.
3. Add `DATABASE_URL`, `DATABASE_AUTH_TOKEN`, `AUTH_SECRET`, `SITE_URL` in Vercel and create a Blob store.
4. Open `https://your-site.vercel.app/admin` and sign in with that email and password.
5. Check `https://your-site.vercel.app/api/health` shows `{"ok":true}`.

The full step-by-step guide is `docs/DEPLOYMENT.md` in the project.

13 • Future sections you can add

All of these fit the existing registry pattern (one definition + one component) and can be switched on/off like today's sections.

GitHub & coding stats

Live contribution graph, top languages, LeetCode/Codeforces/Kaggle ranks pulled automatically.

Built-in blog / notes

Full articles written in the admin, with their own SEO pages, reading time and tags.

Publications & research

Papers, posters, reports with PDF links and citations.

Open-source contributions

PRs and repos you contributed to, with stars and merged status.

Hackathons

Event cards with team, problem statement, result and demo video.

Courses & learning log

Every course (NPTEL, Coursera, IITM) with progress and notes.

Services / hire me

What you can do for clients, with packages and an enquiry form.

Now page

'What I'm doing now' — current course, book, project, location.

Uses / setup

Your laptop, tools, VS Code extensions and desk setup.

Volunteering & leadership

Separate section if Experience gets long.

Media & press

Interviews, features, podcasts, newspaper mentions.

Resume timeline (interactive CV)

One-page interactive CV with print-to-PDF button.

AI 'Ask me anything'

A chat assistant trained on your portfolio content that answers recruiter questions.

Guestbook

Visitors leave a short note (with moderation queue in admin).

FAQ

Common recruiter questions: availability, location, relocation, notice period.

Language & hobbies

Languages spoken, sports, music, photography.

Platform upgrades (Phase 4–5)

Upgrade	Benefit
Per-section layout switcher	Pick any of the six designs per section from the admin
Theme presets	Neon / Netflix red / Gold film / Light in one click
Email alerts (Resend)	Get notified about new messages
Two-factor login (TOTP)	Stronger admin security
Privacy-friendly analytics	See visits and popular projects without cookies

14 • Folder structure & how to extend

```

cinematic-portfolio/
├─ src/
│  ├─ app/
│  │  ├─ (site)/      public pages: home, projects/[slug], privacy
│  │  ├─ admin/       login + (panel): dashboard, sections, media, messages, settings, account
│  │  ├─ api/         auth, admin CRUD, contact, health
│  │  ├─ media/[...path] safe serving of local uploads
│  │  └─ layout.tsx · not-found.tsx · error.tsx · global-error.tsx · robots.ts · sitemap.ts
│  ├─ components/
│  │  ├─ site/        Navbar, Footer, Preloader, Galaxy, Cursor, sections/*.tsx
│  │  ├─ admin/       AdminShell, SectionEditor, DynamicForm, MediaPicker, ...
│  │  └─ ui/          Modal, Markdown, States, SocialIcon
│  ├─ lib/
│  │  ├─ registry.ts  ★ section types + fields (edit here to add fields)
│  │  ├─ settings-def.ts ★ site settings groups
│  │  ├─ validation.ts Zod builders, URL safety, slugify
│  │  └─ server/      auth, session, http wrapper, rate-limit, storage, content queries
│  ├─ db/            schema, migrations, demo seed data
│  └─ proxy.ts       CSP nonce + admin guard
├─ scripts/         setup.ts, reset-password.ts, gen_demo_images.py
├─ public/demo/     placeholder artwork
├─ docs/            PLAN, DEPLOYMENT, PRODUCTION_PAGE_AUDIT, this guide
└─ tests/           unit tests

```

Add a new field to an existing section

1. Open `src/lib/registry.ts` and add e.g. `{ name: "gpa", label: "GPA", type: "text" }` to Education's `itemFields`.
2. Show it in `src/components/site/sections/Education.tsx` with `str(d.gpa)`.
3. Done — the admin form and validation update automatically.

Add a brand-new section

1. Add a type to `SECTION_TYPES` in the registry.
2. Create its component in `components/site/sections/` and register it in `SectionRenderer.tsx`.
3. Add a row for it in `db/seed-data.ts` (or insert it once with the setup script).
4. Turn it on in Admin → Sections.

15 • Master prompt (how this portfolio was specified)

Use this prompt with an AI coding assistant to rebuild the project from scratch, or as a brief for a developer.

You are a senior full-stack engineer, UI/UX designer and security engineer.
Build a production-ready, cinematic 3D personal portfolio with a secure admin CMS for <NAME>, a BS Data Science & Applications student at IIT Madras.

STACK: Next.js 16 (App Router, TypeScript), React 19, Tailwind CSS 4, Three.js with React Three Fiber + drei, GSAP ScrollTrigger, Framer Motion, Lenis smooth scroll, Drizzle ORM + libSQL (SQLite file locally, Turso in the cloud), jose JWT sessions, bcrypt, Zod, optional Vercel Blob for uploads, Vitest for tests.

DESIGN: dark sci-fi neon. Background #05040b, accents violet #8b5cf6 and cyan #22d3ee (editable in admin). Glass cards, gradient text, Space Grotesk / Inter / JetBrains Mono (self-hosted). One big motion moment per section; respect prefers-reduced-motion; lazy-load all 3D; fewer particles on mobile.

PUBLIC SITE – 13 sections, each can be enabled/disabled, reordered, renamed and shown/hidden in the menu from the admin:

- 1 Hero: name reveal, typing roles, draggable 3D photo card (front/back), rings, galaxy particle background, CTA + résumé button, socials.
- 2 About: tilt photo, Markdown bio, fact chips, animated counters, interests.
- 3 Education: scroll-drawn timeline, Completed/Ongoing/Upcoming badges.
- 4 Skills: 3D tag sphere + category tabs with proficiency bars.
- 5 Projects: featured layout, filters, search, status badges, show-more, case-study page per project (/projects/[slug]) with video and gallery.
- 6 Experience
- 7 Certifications (image/PDF viewer, verify link)
- 8 Achievements (GSAP pinned horizontal film reel)
- 9 Gallery (3D ring + lightbox)
- 10 Coming Soon / roadmap with progress
- 11 Testimonials marquee
- 12 Writing / blog links
- 13 Contact form → admin inbox.

Global: cinematic preloader, custom cursor, film grain, offline banner, SEO metadata, JSON-LD, robots + sitemap, 404/500/maintenance pages.

ADMIN (/admin): login (no public sign-up), dashboard with launch checklist, sections manager (switches + order), generic entry editor generated from ONE section registry (fields: text, textarea, markdown, url, date, number, select, tags, boolean, image, images, file, color), media library with drag-drop, browser-side image compression and alt text, messages inbox, site settings (profile, socials, colours/effects, SEO, footer, maintenance, privacy), account page (change password, sign out everywhere, audit log).

SECURITY: bcrypt(12); rate-limit + lockout; no account enumeration; httpOnly SameSite cookie with session version for revocation; server-side authorization on every API; Origin check for CSRF; nonce-based CSP; Markdown without raw HTML; URL allow-list (block javascript:, //host); uploads checked by magic bytes, no SVG, random names, safe serving; safe post-login redirects; generic 500s with reference IDs; secrets only in .env.local.

UX STATES: loading, empty, no-results, error with retry, success, offline, session-expired, maintenance; keep form input after errors; accessible dialogs, labels, focus rings, skip link.

DELIVER: `npm run setup` (DB + demo data + admin), README, deployment guide (Vercel + Turso + Blob, Render, VPS), unit tests, lint/typecheck clean.

Upgrade prompt templates

Replace the <angle-bracket> parts and paste into Claude in this project.

Add a new section type

Add a new section type called "<Hackathons>" to the section registry (src/lib/registry.ts) with fields <title, event, date, result, team, image, link>. Create a public component in src/components/site/sections/, register it in SectionRenderer, add demo seed data, and make sure it can be switched on/off and reordered from the admin. Keep the existing design system.

Switch a section's design

Redesign the <Projects> section as <Netflix-style horizontal rows per category with hover-expanding cards>. Keep the same registry fields and admin controls. Add a section setting "layout" (select: grid | rows) so I can switch between the old and new design from the admin.

Add a feature to the admin

Add email notifications for new contact messages using Resend. API key from env RESEND_API_KEY; if missing, skip silently. Add a setting toggle in Admin → Settings → Notifications. Never expose the key to the browser.

Add a theme preset

Add theme presets (Neon, Netflix red, Gold film, Light minimal) to Admin → Settings → Appearance that set accent colours, fonts and effect toggles in one click.

16 • What to send for your real version

Basics

Full name & logo initials · headline · one-line tagline · public email · city · LinkedIn, GitHub, LeetCode, Kaggle, Instagram links · résumé PDF · 3–5 roles for the typing animation.

Photos

2 portrait photos for the 3D card (vertical 4:5, ≥ 1200 px tall, good light) · 1 photo for About · event/campus photos for the gallery.

Education

Each level: institution, programme, dates, status (completed / ongoing / upcoming), grade, key courses.

Skills

List with level (beginner / intermediate / advanced) and category.

Projects

Title, summary, what you did, tech, GitHub/live/video links, screenshots, status (done / in progress / planned).

Certificates & achievements

Title, issuer, date, credential ID/URL, image or PDF · awards with dates and photos.

Experience & future

Internships, clubs, volunteering · upcoming courses, project ideas and goals with target dates.

Decisions

Domain name (or free .vercel.app) · which sections to hide for now · privacy contact email and message retention period.

The privacy notice is a template. Fill in your real details and have it reviewed if unsure — this guide does not claim legal compliance (GDPR / India DPDP etc.).

Commands cheat-sheet

```
npm install                # first time
npm run setup              # database + demo content + admin login
npm run dev               # http://localhost:3000 · admin: /admin
npm run setup -- --reset   # reload demo content (keeps admin + uploads)
npm run admin:reset-password -- you@example.com "NewPass2026"
npm run build && npm start  # production mode
npm test · npm run lint · npm run typecheck
```